# Seongju Lee

Ph.D. in AI Convergence


South Korea

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## SUMMARY


Senior Researcher at HD Hyundai Robotics working on robot learning and simulation for humanoid and dual-arm robots. I build teleoperation systems, multimodal demonstration pipelines, and environments for testing learned policies. My research background combines robust visual perception and contrastive representation learning with experience integrating AI models into autonomous systems.


## SKILLS

**Robot Learning:** Imitation Learning, VLA, WAM
**Simulation:** Isaac Sim, MuJoCo
**Perception:** Detection, Segmentation, Tracking, 6D Object Pose Estimation
**Learning:** Supervised Contrastive / Metric Learning, Multimodal Learning, Domain Generalization, Knowledge Distillation
**Programming:** Python, C++, C, MATLAB
**Frameworks:** PyTorch, ROS/ROS 2, LeRobot
**Robot Platforms:** UR5, Franka Panda, HD Hyundai Robotics dual-arm robot
**Languages:** Korean (Native), English

## RESEARCH EXPERIENCE

Jul. 2026 – Present — **Senior Researcher** — **HD Hyundai Robotics**
Developing robot learning pipelines and simulation environments for intelligent dual-arm robots.

Mar. 2026 – Jun. 2026 — **Postdoctoral Researcher** — **GIST Institute of Artificial Intelligence**

## SELECTED PUBLICATIONS

### Published Papers (*: Equal Contribution)

Journal metrics: 2025 JIF and best JCR subject-category quartile. Full publication list: Google Scholar.

Top Conference

**MART: MultiscAle Relational Transformer Networks for Multi-agent Trajectory Prediction**
The 18th European Conference on Computer Vision (ECCV 2024), Poster
**TL;DR**: Graph Neural Networks (GNNs), Hypergraph Transformer
**Seongju Lee** (First author), Junseok Lee, Yeonguk Yu, Taeri Kim, and Kyoobin Lee
[Paper] [Code] 27.9% Acceptance Rate Cited by 80

Conference

**MV2: A Large-Scale 360-degree Multi-View Maritime Vision Dataset for Object Detection and Segmentation**
The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
Junseok Lee*, Jongwon Kim*, **Seongju Lee*** (Co-first author), Taeri Kim, and Kyoobin Lee
[Paper] [Website]

Top Conference

**ManipForce: Force-Guided Policy Learning with Frequency-Aware Representation for Contact-Rich Manipulation**
2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
**TL;DR**: Imitation Learning, Multimodal Policy, Vision Foundation Models (VFMs), Force/Torque Sensor
Geonhyup Lee, Yeongjin Lee, Kangmin Kim, **Seongju Lee**, Sangjun Noh, Seunghyeok Back, Kyoobin Lee
[Paper] [Website] [Code] 38.04% Acceptance Rate Cited by 29

**Selected Publications (continued)**

Top Conference

**CD-FKD: Cross-Domain Feature Knowledge Distillation for Robust Single-Domain Generalization in Object Detection**

2026 IEEE International Conference on Robotics and Automation (ICRA 2026)

**TL;DR**: Knowledge Distillation, Robust Object Detection, Single-Domain Generalization

Junseok Lee*, Sungho Shin*, **Seongju Lee**, Kyoobin Lee

[Paper] [Code] 38.04% Acceptance Rate Cited by 1

Conference

**Robust Maritime Object Detection under Adverse Conditions via Joint Semantic Learning without Extra Computational Overhead**

The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)

Junseok Lee, **Seongju Lee**, Jongwon Kim, Jumi Park, and Kyoobin Lee

[Paper] 46% Acceptance Rate Cited by 2

Top Conference

**Block Selection Method for Using Feature Norm in Out-of-Distribution Detection**

2023 IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR 2023), Poster

Yeonguk Yu, Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee

[Paper] [Code] 25.8% Acceptance Rate Cited by 82

Journal

**Human-efficient annotation system via active semi-supervised learning and selective review**

Intelligent Systems with Applications, Vol. 31, Article 200701, 2026

Jongwon Kim, Youmin Park, Seunghyeok Hyun, Yunjae Heo, **Seongju Lee**, Yeonguk Yu, and Kyoobin Lee

[Paper] JCR 2025 Q1

## Additional First-/Co-first-Author Publications

Journal

**SleePyCo: Automatic Sleep Scoring with Feature Pyramid and Contrastive Learning**

Expert Systems with Applications, 2024

**Seongju Lee** (First author), Yeonguk Yu, Seunghyeok Back, Hogeon Seo, and Kyoobin Lee

[Paper] [Code] JIF 9.4 (2025, Q1) Cited by 107

Journal

**Robust Skin Disease Classification by Distilling Deep Neural Network Ensemble for the Mobile Diagnosis of Herpes Zoster**

IEEE Access, 2021

Seunghyeok Back*, **Seongju Lee*** (Co-first author), Sungho Shin, Yeonguk Yu, Taekyeong Yuk, Saepomi Jong, Seungjun Ryu, Kyoobin Lee

[Paper] JIF 4.2 (2025, Q2) Cited by 80

Journal

**Intra- and inter-epoch temporal context network (IITNet) using sub-epoch features for automatic sleep scoring on raw single-channel EEG**

Biomedical Signal Processing and Control, 2020

Hogeon Seo*, Seunghyeok Back*, **Seongju Lee*** (Co-first author), Deokhwan Park, Tae Kim, Kyoobin Lee

[Paper] [Code] JIF 5.7 (2025, Q2) Cited by 313

## Manuscript in Submission

In Submission

**Rethinking Deformable Convolution as an Adapter with Cross-layer Weight Sharing for Robust Semantic Segmentation in the Wild**

Target Journal: IEEE Robotics and Automation Letters

First Prize in the USV-based Obstacle Segmentation Challenge at the 3rd Maritime Computer Vision Workshop

**TL;DR**: Vision Foundation Models (VFMs), Domain Generalized Semantic Segmentation (DGSS), Deformable Convolution Networks (DCNs), Parameter-Efficient Fine-Tuning (PEFT)

**Seongju Lee** (First author), Raeyoung Kang, Sangbeom Lee, Sangjun Noh, Yeonguk Yu, Junseok Lee, and Kyoobin Lee

## SELECTED PROJECTS

Jul. 2026 – Present
AI & Software Developer

**Development of an Intelligent Dual-Arm Robot**
HD Hyundai Robotics
**TL;DR**: Robot learning with VLA/WAM models; VR/glove teleoperation and hand retargeting; simulation and multimodal demonstration data pipelines.
[News]

May 2025 – Feb. 2026
AI Developer

**Development of Tactile Intelligence in Robotic Hands Based on Tactile Data Learning for Manipulating Irregular Multiple Types of Objects**
Ministry of Trade, Industry and Energy (MOTIE), South Korea
**TL;DR**: Multimodal Action Policy for Contact-Rich Tasks using Vision and F/T Sensing

Sep. 2023 – Feb. 2026
Project Member
Software Developer

**Development of a Micro Intelligent Drone for Autonomous Missions in Communication/GPS-denied Environments**
Institute of Civil-Military Technology Cooperation (ICMTC), South Korea
**TL;DR**: RGB-IR object detection and a multimodal RGB, IR, and LiDAR database captured from drone views.

Apr. 2023 – Dec. 2024
Project Member
Software Dev. Leader

**Development of a Situational Awareness System for Preventing Collisions and Accidents of Autonomous Ships**
Ministry of Trade, Industry and Energy (MOTIE), South Korea
**TL;DR**: Multi-camera maritime object detection and tracking, training dataset creation, and integration into autonomous ship systems.

Apr. 2020 – Nov. 2024
Project Leader
AI & Software Dev. Leader

**Development of Swarm USV Autonomous Navigation Algorithm**
Agency for Defense Development (ADD), South Korea
**TL;DR**: Maritime object detection and intention prediction for swarm situational awareness, integrated into multiple USVs. [News]

Mar. 2022 – Dec. 2022
Project Member
AI & Software Dev. Leader

**Ceiling Layout Estimation for Indoor Environments**
LG, South Korea
**TL;DR**: Ceiling segmentation and dataset construction for robust floor plan generation under occlusion.

## EDUCATION

Mar. 2020 – Feb. 2026

**Gwangju Institute of Science and Technology (GIST)** **Advisor: Prof. Kyoobin Lee**
Ph.D. in AI Convergence
Dissertation Title: Robust Perception and Trajectory Forecasting for Autonomous Navigation in the Wild

Mar. 2018 – Feb. 2020

**Gwangju Institute of Science and Technology (GIST)**
M.S., School of Integrated Technology (Intelligent Robotics Program)

Mar. 2014 – Feb. 2018

**Gwangju Institute of Science and Technology (GIST)**
B.S. in Mechanical Engineering

## SELECTED AWARDS

Jun. 2026

**Multimodal Semantic Segmentation Challenge, 4th Workshop on Maritime Computer Vision (MaCVi)** **CVPR 2026**
First Place

Jan. 2026

**RoCo Challenge @ AAAI 2026: Robotic Collaborative Assembling for Human-Centered Manufacturing** **AAAI, A*STAR, NTU**
Finalist Excellence Award; integrated 6D pose estimation (MegaPose + ICP).

Nov. 2025

**Autonomous Driving AI Challenge (Semantic Segmentation)** **MSIT, IITP, KADIF**
Second Prize (IITP President's Award)

Dec. 2024

**USV-based Obstacle Segmentation Challenge, MaCVi 2025** **WACV 2025**
First Prize

Dec. 2024

**USV-based Panoptic Segmentation Challenge, MaCVi 2025** **WACV 2025**
Second Prize

Sep. 2022

**The first NIA-GIST AI-HUB AI Competition** **NIA, GIST**
First Prize